

Toast and Butter Service SOP

One episode toasts two slices of bread, butters them, plates them, and puts the plate on the delivery tray. The two slices come off the bread rack, go into the toaster, run one cycle, land flat on the board, get buttered, go onto the serving plate side by side, and the plate is carried onto the tray. The knife is back in its spot and both arms are home when the episode ends.

The order never changes. Near slot first, far slot second, and that order carries through the whole episode: the near slice is toasted first, retrieved first, buttered first, and plated first.

Four rules hold for the whole episode. Nothing but bread goes into a slot. No gripper touches the toaster while the cycle runs. One cycle per episode, and the lever is never forced up early. Nothing is passed hand to hand: every item that changes arms does so by sitting on a surface first.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the bread rack on the right, the toaster at the back, the knife spot, the board in the middle, the butter tub on the left, the plate spot, and the delivery tray at the front edge.
3. Both arms are at home with grippers open.
4. The tabletop is clear of anything but the zones listed below.
5. Nothing blocks the paths between the bread rack, the toaster, the board, the plate, and the tray.

Materials checklist

1. The toaster stands at the back of the tabletop, plugged in, cold, and empty, with its lever up. Its two slots run left to right, so there is a **near slot** and a **far slot**. The lever is on the right side of the toaster, within reach of the right gripper.
2. The browning dial is set to the same middle setting for every episode. It is not touched during an episode.
3. The toaster lifts the finished slices high enough that a good part of each slice stands above its slot, so a gripper can take the top edge without reaching into the slot.
4. The toaster crumb tray is in place and empty.
5. The bread rack sits against the right edge and holds two slices of bread standing on edge, their faces parallel to the toaster slots, with the tops clear so a gripper can take each by its

top edge. The two slices are the same size, dry, and not torn.

6. The knife lies flat in its knife spot, on the clear patch between the bread rack and the board, with the whole handle clear of anything else so a gripper can take it from above. There is no rest, cradle, or slot: the knife is simply set down. Its handle is bulky enough to grasp off a flat surface. The blade is clean and dry.
7. The butter tub sits open on its tub spot at the back left, lid off and set aside off the tub spot. The butter is soft enough to take onto the blade without scraping, and the tub holds enough to cover two slices with room to spare.
8. The board lies flat in the middle of the tabletop and is bare.
9. The serving plate sits empty and right side up on the plate spot, in front of the board.
10. The delivery tray lies flat at the front edge, empty, directly in front of the plate spot.

Workspace layout

- **Bread rack:** against the right edge, within reach of the right arm. It holds the two slices standing on edge and is empty at the end of the episode.
- **Toaster:** at the back of the tabletop, in the middle, reachable by both arms. The right arm works the lever and the slots. The left arm only steadies the body.
- **Knife spot:** the clear patch between the bread rack and the board, within reach of the right arm. The knife lies flat here whenever it is not in a gripper. It is a bare patch, not a rest or a holder.
- **Board:** flat in the middle of the tabletop. Both slices are laid here and buttered here. It is bare at the start and holds nothing but crumbs at the end.
- **Butter tub:** open on its tub spot at the back left, close enough to the middle that the right gripper reaches it without stretching and the left gripper reaches its rim.
- **Plate spot:** in front of the board, in the middle of the tabletop, reachable by both arms.
- **Delivery tray:** flat at the front edge, straight in front of the plate spot, so the plate travels toward the front edge and not sideways.

Ownership rule, stated once. The right arm owns the bread, the toaster lever, the knife, and every slice move. The left arm steadies the toaster and holds the butter tub down. Both grippers share the plate, and only for the carry onto the tray. Neither arm takes over the other's work.

Vocabulary

- **Arm assignment:** the right gripper moves every slice, works the lever, and holds the knife. The left gripper steadies the toaster and holds the butter tub down, and it never picks up a slice, never takes the knife, and never works the lever. Both grippers take the plate rim in Step 6.
- **Near slot and far slot:** the toaster slot nearer the front edge is the near slot. The one behind it is the far slot. The near slot is always filled and emptied first.

- **Steady hold:** the left gripper, closed and empty, resting against the left side of the toaster body so the toaster cannot shift. It rests against the body and takes no grip on it.
- **Cycle done:** the lever has sprung back up on its own and both slices stand above their slots.
- **Abnormal toaster condition:** smoke, a burning smell, flame, or a toaster that shifts off its spot while the cycle runs.
- **Loaded blade:** butter sits along the flat of the blade, enough to draw across a slice without going back to the tub after one pass.
- **Full coverage:** the whole top face of a slice carries butter, corner to corner and out to the crust, with no bare bread showing.
- **Buttered face:** the face of the slice that was buttered. It faces up from the moment it is buttered until the episode ends.
- **Stable placement:** the item stays still after the gripper opens, and stays still for 2 seconds.
- **Release point:** the only places a gripper may open. They are a toaster slot, the board, the serving plate, the knife spot, and the tray for the plate. Never open a gripper over open space, over the butter tub, or over the bread rack.
- **Tool home:** the knife spot. The knife lies flat there whenever it is not in a gripper.

Steps

Step 1: Load the two slices

Goal: both slices stand in the toaster, near slot first, and the toaster has not moved.

- The **left gripper** takes the steady hold on the left side of the toaster body and keeps it for the whole step.
- The **right gripper** grips the first slice by its top edge, lifts it straight up clear of the bread rack, and carries it level to the toaster.
- Center it over the **near slot**, lower it straight down until it rests in the slot, then release.
- The **right gripper** takes the second slice the same way and lowers it into the **far slot**.
- One slice per trip. Never carry both.
- Keep each slice standing on edge the whole way. Do not turn the wrist on this carry.
- The **left gripper** lifts clear once both slices are in.

Check: both slices stand down in their slots, the near slot was filled first, and neither slice leans out or rests across the mouth of a slot. The toaster is still on its spot. If a slice sits across the slot mouth instead of dropping in, the **right gripper** lifts it clear by the top edge and lowers it again.

Step 2: Run one toasting cycle

Goal: one full cycle runs from start to finish with both arms clear of the toaster.

2.1 Start the cycle

- The **left gripper** takes the steady hold on the left side of the toaster body.
- The **right gripper** presses the lever straight down until it latches and stays down, then releases and moves clear.
- The **left gripper** lifts clear once the lever is latched.

2.2 Wait out the cycle

- Both arms stay clear of the toaster while it runs. Do not touch the toaster, the lever, the dial, or the slices.
- Wait until the lever springs back up on its own and both slices stand above their slots.
- One cycle per episode. Never push the lever down a second time, and never force it up early.

Abnormal condition branch: if smoke, a burning smell, or flame appears, or the toaster shifts off its spot, the **right gripper** lifts the lever straight up to end the cycle. Both arms then move clear. Do not retrieve the slices and do not butter them. End the episode unsuccessful.

Check: the lever is up, both slices stand above their slots, and the toaster is still on its spot. If the lever stays down and nothing rises, do not press it, pull it, or restart the toaster. End the episode and log a toaster fault.

Step 3: Lift the slices onto the board

Goal: both slices lie flat and apart on the board and the toaster is empty.

- The **left gripper** takes the steady hold on the left side of the toaster body and keeps it for the whole step.
- The **right gripper** grips the near slice by the top edge that stands above the slot and lifts it straight up out of the slot. Lift straight up. Do not drag the slice against the mouth of the slot.
- Turn the wrist while carrying, clear of the board, so the slice comes level. Never shift the slice inside the fingers to do it.
- Lower it flat onto the right half of the board and release once it is resting.
- The **right gripper** takes the far slice the same way and lays it flat on the left half of the board, not touching the first slice.
- One slice per trip. Never lift both.
- The **left gripper** lifts clear once both slices are on the board.

Check: both slices lie flat and apart on the board, neither hangs over an edge, and both slots are empty. The toaster is still on its spot. If a slice lands leaning or overhanging, the **right**

gripper lifts it clear by an edge and lays it down again.

Expected state: the bread rack is empty, the toaster is empty with its lever up, the board holds two slices lying flat, and the knife is lying in its knife spot.

Step 4: Butter both slices

Goal: the top face of each slice is covered out to the crust and the knife is back in its spot.

Butter the right slice first, then the left slice.

4.1 Load the knife

- The **right gripper** takes the knife by the butt of the handle, the end furthest from the blade, and lifts it straight up off the knife spot with the blade flat.
- The **left gripper**, closed and empty, comes down on the rim of the butter tub and holds the tub down against the tabletop. It takes this hold **before the blade goes in**.
- The **right gripper**, keeping the blade flat, lowers it into the tub, presses it into the butter, and lifts it straight out.
- The **left gripper** lifts clear of the tub only once the blade is out and clear of the rim.

The tub is light and the blade drags on the way out, so an unheld tub lifts with the knife or slides off its spot. Anchor it every time, including every reload.

4.2 Draw the butter across

- The **right gripper** carries the knife level to the board, blade still flat, and lays it on the near edge of the slice so the **flat of the blade** meets the toast.
- Draw the blade from the near edge to the far edge, keeping it flat.
- Work across the slice in passes until the whole top face is covered, corner to corner and out to the crust, with no bare bread showing.
- Lift the blade clear of the slice before moving away.

Full coverage is the standard, not a pass count. Take as many passes as the slice needs.

If the knife runs dry before the face is covered, the **right gripper** carries it back through Step 4.1, tub hold included, and carries on where it stopped. Keep hold of the knife between loads.

Keep the butter on the toast. Do not run the blade off the crust onto the board.

4.3 Butter the left slice

- Reload the knife through Step 4.1, tub hold included.
- Butter the left slice the same way, near edge to far edge, until the whole top face is covered.

4.4 Return the knife

- The **right gripper** carries the knife back to the knife spot, lays it down flat, and releases.

Check: both slices carry butter over the whole top face out to the crust with no bare patches, the knife lies flat in its knife spot, the butter tub is upright on its tub spot, no butter has been dragged onto the board, and both slices are still flat and apart. If a bare patch is left, the **right gripper** takes the knife back up and covers it before moving on. If a slice was pushed out of place, the **right gripper** lifts it clear and lays it down again. If the tub has been pushed off its spot, the **left gripper** sets it back before the next load.

Step 5: Plate the two slices

Goal: both slices lie flat on the plate, side by side, buttered faces up.

Take the right slice first.

- The **right gripper** grips the slice by the crust on its right edge, lifts it straight up off the board, and carries it level to the plate.
- Keep the slice level so the buttered face stays up. Do not turn the wrist on this carry.
- Lower it flat onto the far half of the plate and release once it is resting.
- The **right gripper** takes the left slice the same way and lays it on the near half of the plate, beside the first slice and not on top of it.
- One slice per trip. Never lift both.

Check: both slices lie flat on the plate, buttered faces up, side by side rather than stacked, and neither overhangs the rim. If a slice lands buttered face down or off the plate, the **right gripper** lifts it clear and lowers it again.

Expected state: the plate holds both buttered slices. The board holds nothing but crumbs, the knife lies in its knife spot, and the toaster and bread rack are empty.

Step 6: Deliver the plate to the tray and end the episode

Goal: the plate rests flat on the delivery tray and both arms are home.

6.1 Carry the plate onto the tray

- The **right gripper** takes the plate rim on its right side and the **left gripper** takes the rim on its left side, the two holds opposite each other.
- Both grippers lift the plate level, just clear of the tabletop.
- Carry it straight toward the front edge and onto the tray, keeping it level so nothing slides.
- Lower it until it rests flat on the tray, then release the **left gripper**, then the **right gripper**.
- Do not slide or drag the plate across the tabletop, and do not carry it with one gripper.

Check: the plate sits flat in the middle of the tray with no part of it over the tray edge, and both slices still lie flat on it with their buttered faces up. If the plate lands off the tray or across its edge, both grippers take the rim again, lift it clear, and lower it again.

6.2 End the episode

- Confirm the plate is on the tray with both buttered slices on it, the knife is lying flat in its knife spot, and the toaster is empty with its lever up.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the plate off the tray and set the two slices aside for disposal.
2. Wipe the plate clean and dry and set it back empty and right side up on the plate spot.
3. Brush the crumbs off the board, wipe it clean and dry, and put it back flat in the middle of the tabletop.
4. Wipe the knife clean and dry and lay it flat on its knife spot.
5. Pull the toaster crumb tray, empty it, and put it back. Check the toaster is cold, empty, and its lever is up.
6. Check the browning dial is still on the same middle setting.
7. Stand two fresh slices on edge in the bread rack with their faces parallel to the toaster slots. Vary how the slices sit between episodes rather than rebuilding the same layout every time.
8. Replace any slice that is torn, stale, or dried hard.
9. Top up the butter tub if it no longer holds enough to cover two slices, and leave it open on its tub spot with the lid off.
10. Wipe the tray and the tabletop, then run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Slices loaded in the wrong order or the wrong way

- **Visible cue:** the right gripper fills the far slot first, carries both slices in one trip, opens above a slot and lets a slice fall in, or leaves a slice lying across the mouth of a slot and moves on.
- **SOP rule broken:** Step 1, near slot first and far slot second, one slice per trip, lowered straight down until it rests in the slot before the right gripper releases.
- **Coaching note:** near slot first, one slice per trip, all the way down before you open.

Violation: Toaster not steadied by the left gripper

- **Visible cue:** the right gripper loads a slice, presses the lever, or lifts a slice out while the left gripper is away from the toaster body, or the left gripper grips the toaster instead of resting against it. The toaster rocks or slides on its spot.
- **SOP rule broken:** Steps 1, 2.1, and 3, the left gripper rests closed and empty against the left side of the toaster body for the whole of each of these steps and takes no grip on it.
- **Coaching note:** the left arm goes on the toaster before the right arm does anything to it.

Violation: Cycle not run as one full cycle

- **Visible cue:** the lever is pressed without latching, forced back up before it springs on its own, pressed down a second time, or the browning dial is turned during the episode. The cycle also runs on after smoke or a burning smell appears, or the slices are retrieved and buttered after an abnormal condition.
- **SOP rule broken:** Step 2, press the lever until it latches, one cycle per episode, never force it up early, and on an abnormal condition lift the lever with the right gripper, clear both arms, and end the episode unsuccessful.
- **Coaching note:** one cycle, start to finish. Stop only for smoke or flame, and then stop for good.

Violation: Toaster touched while the cycle runs

- **Visible cue:** either gripper touches the toaster body, the lever, the dial, or a slice between the lever latching and the lever springing back up.
- **SOP rule broken:** Step 2.2, both arms stay clear of the toaster while it runs.
- **Coaching note:** hands off from the latch until the pop.

Violation: Toast dragged out or taken by the wrong part

- **Visible cue:** the right gripper reaches down into the slot, grips a slice by its face instead of the top edge that stands above the slot, drags the slice up against the mouth of the slot, lifts the far slice first, or lifts both slices in one trip. The right gripper also works the slice around inside its fingers to lay it flat, or turns the wrist only once the slice is already touching the board.
- **SOP rule broken:** Step 3, grip the near slice first by the top edge standing above the slot, lift it straight up out of the slot, one slice per trip, and turn the wrist while carrying and clear

of the board rather than shifting the slice inside the fingers.

- **Coaching note:** take the part you can see, lift it straight up, and turn it on the way over.

Violation: Butter tub not held, or gripped and moved

- **Visible cue:** the blade goes into the tub with no left gripper on it, the left gripper arrives after the blade is already in, or it lifts away while the blade is still in the butter. The left gripper also closes on the tub and picks it up or drags it during the hold. The tub lifts with the knife, slides, or tips.
- **SOP rule broken:** Step 4.1, the left gripper holds the tub down against the tabletop before the blade goes in and lifts clear only once the blade is out and clear of the rim, on every load including reloads, and it holds the tub down without taking hold of it.
- **Coaching note:** anchor first, release last, and hold it down rather than pick it up.

Violation: Knife regripped in the air or left off its spot

- **Visible cue:** the right gripper shifts its hold along the handle without setting the knife down, the knife is parked on the board or the plate between uses, or it is still in a gripper while a slice is moved.
- **SOP rule broken:** Steps 4.1 and 4.4, take the knife by the butt of the handle off the knife spot, and lay it back flat in the knife spot when the buttering is done.
- **Coaching note:** the knife goes home before anything else moves.

Violation: Butter not kept on the face of the slice

- **Visible cue:** the knife is put down with bare bread still showing, butter stopping short of the crust on one or more edges, or a gap between passes left uncovered. The blade also runs off the crust and smears butter onto the board, or a dry blade is dragged over the toast instead of going back to the tub.
- **SOP rule broken:** Steps 4.2 and 4.3, work across the slice in passes until the whole top face is covered out to the crust, keep the butter on the toast, and reload through Step 4.1 when the knife runs dry.
- **Coaching note:** check the whole face before you put the knife down, and stop each stroke at the crust.

Violation: Slice pressed, nudged, or slid instead of lifted

- **Visible cue:** a gripper pushes, drags, or shoves a slice across the board or the plate to line it up, or presses, pats, or flattens a slice after releasing it, instead of lifting it clear and lowering it again.
- **SOP rule broken:** Steps 3, 4, and 5, correct a bad placement by lifting the slice clear with the right gripper and setting it down again, and never press or flatten a slice.
- **Coaching note:** lift it and place it again, or leave it alone.

Violation: Slices plated wrong

- **Visible cue:** a slice lands buttered face down, the two slices are stacked on each other, a slice overhangs the rim, the left slice is plated first, or both slices are carried in one trip.
- **SOP rule broken:** Step 5, take the right slice first, one slice per trip, and lay both slices flat and side by side on the plate with their buttered faces up and nothing overhanging the rim.
- **Coaching note:** right slice first, butter side up, side by side.

Violation: Plate delivered wrong

- **Visible cue:** the plate is carried or lifted with one gripper, tilted far enough that a slice slides, dragged or slid across the tabletop, released above the tray, or left across the tray edge or off the tray.
- **SOP rule broken:** Step 6.1, both grippers take the plate rim on opposite sides, lift it level, carry it toward the front edge, and lower it until it rests flat on the tray before releasing.
- **Coaching note:** two grippers, level the whole way, and set it down before you open.

Violation: Wrong arm used

- **Visible cue:** the left gripper picks up a slice, takes the knife, or works the lever; or the right gripper takes the steady hold or the butter tub hold. Both grippers on the plate rim in Step 6 are not this violation.
- **SOP rule broken:** Steps 1 to 6, the right gripper moves every slice, works the lever, and holds the knife, and the left gripper only steadies the toaster and holds the butter tub down.
- **Coaching note:** right arm works, left arm steadies. Only the plate is shared.

Violation: Item dropped, collided, or knocked over

- **Visible cue:** a slice falls off the board, the plate, or the tray; the knife strikes the toaster or the plate; the butter tub is knocked over; the bread rack or the toaster is knocked off its spot; or the plate tips in the carry.
- **SOP rule broken:** Steps 1 to 6, keep each item controlled on a clear travel path and release only after a stable placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the plate still on the plate spot, a slice still on the board, the knife left off its knife spot, a slice still in the toaster, or an arm away from home.
- **SOP rule broken:** Step 6.2, confirm the plate on the tray with both buttered slices, the knife flat in its knife spot, and the toaster empty with its lever up; return both arms home; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Toaster fault:** a lever that will not latch, a lever that stays down and never springs back, a cycle that ends with the slices still sitting low in the slots, or a toaster that will not heat. Check the toaster before the next episode.
- **Defective bread:** a slice found torn, stale, or dried hard mid episode, or one that breaks apart when it is lifted. Replace it before the next episode.
- **Butter tub runs empty** before both faces are covered, leaving nothing left to load the knife with. Top it up before the next episode.
- **Knife fault:** a handle too slick or too slim for the gripper to lift off a flat surface. Replace the knife before the next episode.
- **Board, plate, or tray shifts out of its spot** during the episode.

Annotation subtasks (from SOP)

1. Steady the toaster with the left gripper
2. Move one slice from the bread rack into a toaster slot
3. Push the lever down and run one toasting cycle
4. Lift one toasted slice out of the toaster onto the board
5. Hold the butter tub down while the knife loads
6. Take the knife off its spot and load it with butter
7. Draw the butter across one slice
8. Return the knife to its spot
9. Move one buttered slice from the board to the plate
10. Carry the plate onto the delivery tray with both grippers
11. Return both arms home and end the episode